Danil Fedchenko

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Northwestern Economics

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Fields

Research: Econometrics
Teaching: Econometrics

Education

Ph.D., Economics, Northwestern University — (anticipated) 2026
  Dissertation: Essays in Econometrics
  Committee: Federico Bugni (co-chair), Ivan Canay (co-chair), Joel Horowitz

M.A. (*summa cum laude*), Economics, New Economic School — 2020

B.A., Applied Math and Physics, Moscow Institute of Physics and Technology — 2018

Fellowships & Awards

Dissertation University Fellowship, Northwestern University — 2025–2026
Don Patinkin Award for Exceptional Achievements, New Economic School — 2020
Best Master's Thesis Award, New Economic School — 2020
Best TA Award, New Economic School — 2020

Teaching Experience

Teaching Assistant, Northwestern University — 2021-2025
  Econometrics (undergraduate, graduate), Introductory Microeconomics

Teaching Assistant, New Economic School — 2019-2020
  Probability Theory, Statistics, Econometrics, Topics in Econometrics

Research Experience

Research Assistant, Ivan Canay, Northwestern University — 2023
Research Assistant, Eric Auerbach, Northwestern University — 2024

Refereeing

Journal of Political Economy (Micro), Journal of Business & Economic Statistics

Job Market Paper

"The Treatment Effect on a Summary Index: Interpretation, Inference, and Efficiency"

This paper examines the common practice of combining multiple outcomes into summary indices to estimate the effect of an intervention. While the interpretation of such estimates is often intuitive – a weighted average of component-level effects – the weights used by procedures more complex than the simple average can be problematic. Correct inference requires accounting for data-dependent weights when computing standard errors. The resulting correction depends on the variability of the weighting scheme and the magnitude of the true treatment effect, and disappears when the latter is zero. Since none of the commonly used summary index procedures incorporate sources of variability relevant to the estimator's variance, the resulting estimates do not exhibit the high power often attributed to them. Common data manipulations, such as imputing missing values in index components, often produce indices that differ from what they are intended to represent. In such cases, the interpretation of the resulting estimates is no longer that of a weighted sum; instead, it depends on the specific estimation method and the approach used to handle missing data. This study aims to help applied researchers better understand the econometric procedures they use and to assist readers of these studies in interpreting the results.

Other papers

"Partially Identified Rankings from Pairwise Interactions" with Federico Crippa

This paper considers the problem of ranking objects based on their latent merits using data from pairwise interactions. We allow for incomplete observation of these interactions and study what can be inferred about rankings in such settings. First, we show that identification of the ranking depends on a trade-off between the tournament graph and the interaction function: in parametric models, such as the Bradley-Terry-Luce, rankings are point identified even with sparse graphs, whereas nonparametric models require dense graphs. Second, moving beyond point identification, we characterize the identified set in the nonparametric model under any tournament structure and represent it through moment inequalities. Finally, we propose a likelihood-based statistic to test whether a ranking belongs to the identified set. We study two testing procedures: one is finite-sample valid but computationally intensive; the other is easy to implement and valid asymptotically. We illustrate our results using Brazilian employer-employee data to study how workers rank firms when moving across jobs.

Work in Progress

"On the Statistical Properties of CART Split Selection"

This paper studies the construction of classification and regression trees (CART), a key component of random forests. CART is built through recursive partitioning of the covariate space, where each step selects a split that maximizes an objective function known as impurity reduction. I show that this objective is a monotone transformation of the two-sample t-statistic. Consequently, the algorithm partitions the covariate space at the point yielding the largest magnitude of the t-statistic. A similar principle underlies structural break tests in time series, which also divide samples to maximize such statistics. Leveraging this connection, the paper shows that the CART objective tends to infinity for continuous covariates – even those entirely irrelevant to the outcome – while remaining stochastically bounded for discrete irrelevant covariates. This result provides a theoretical argument for CART's well-documented "preference for continuous covariates," the tendency to make more splits along continuous rather than discrete variables. Analogous to solutions used in structural break tests, this behavior can be mitigated by restricting the number of admissible splits for continuous covariates, for instance by excluding splits that produce highly unbalanced subsamples.

Languages

English (fluent), Russian (mother tongue)

References

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